

Draw three right-angled triangles, each with one angle of 30°.

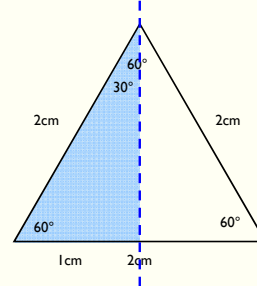
Measure each side to the nearest mm and fill in the table.

triangle	side opposite 30° angle	hypotenuse (side opposite the right angle)
1		
2		
3		

What do you notice about the ratio between the opposite and hypotenuse sides in each triangle?

You should have found that the ratio of **opposite : hypotenuse** is 1:2 or ½.

How can equilateral triangles explain why this is true?



Explore your calculator!

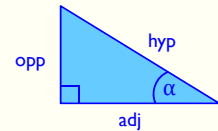
Can you find a **function** which gives the answer ½ when you input 30?

If you input **sin 30** you get ½.

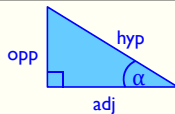
'Sin' is short for sine.

Sine is a function which relates an angle to its opposite and adjacent sides:

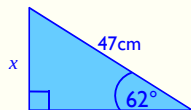
$$\sin \alpha = \frac{\text{opposite}}{\text{hypotenuse}}$$



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We can use sine to calculate a missing length in a triangle:



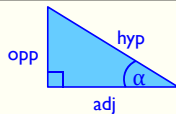
substitute values: $\sin 62 = \frac{x}{47}$

rearrange for x: $x = 47 \times \sin 62$

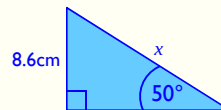
type into your calculator: $x = 41.4985...$

$x = 41.5 \text{ cm}$

$$\sin \alpha = \frac{\text{opposite}}{\text{hypotenuse}}$$



We can use sine to calculate a missing length in a triangle:



substitute values: $\sin 50 = \frac{8.6}{x}$

rearrange for x: $x = \frac{8.6}{\sin 50}$

type into your calculator: $x = 11.2265...$

$x = 11.2 \text{ cm}$

Calculate the length of side x in each of these triangles:

